Perimeter - Mark scheme

Geometry · Edexcel IGCSE Mathematics A (4MA1) Paper 1 · Year 10

Question 1

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · January 2013 · Question 2

2.	$\pi \times 7.6$		2	M1	or $2\times\pi\times\frac{7.6}{2}$
		23.9		A1	for answer which rounds to 23.9
					Total 2 marks

Question 2

Edexcel IGCSE Mathematics A (4MA1) · Paper 1FR · June 2026 · Question 16

Q	Working	Answer	Mark	Notes
	eg $\pi \times 6.3 \left(=\frac{63}{10}\pi\right)$ oe eg $2 \times \pi \times (6.3 \div 2)$ or $2 \times \pi \times 3.15$		2	M1 allow 3.14... or $\frac{22}{7}$ for π
	Correct answer scores full marks (unless from obvious incorrect working)	19.8		A1 accept 19.78 – 19.8
				Total 2 marks

Question 3

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2021 · Question 12

12		$\frac{50}{360}\times\pi\times7\times2$ oe eg $\frac{14\pi}{36}\times5$ **or** "43.98…"÷ 360 × 50 oe		2	M1	Students may use π or 3.14, 3.142 or $\frac{22}{7}$
		Working not required, so correct answer scores full marks (unless from obvious incorrect working)	6.1		A1	Accept answers in the range 6.05 – 6.2
						Total 2 marks

Question 4

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · June 2023 · Question 7

Q	Working	Answer	Mark	Notes
	10 – 3 (= 7) (could be on diagram) **or** 10 + 3 + 10 + 3 (= 26) oe **or** (10 + 3 + 10 + 3) × 3 (= 78) oe **or** 6 × 10 + 4 × 3 (= 72) oe **or** 4 × 10 + 4 × 3 (= 52) oe		3	M1 for 10 – 3 (= 7) (could be on diagram) or for finding the perimeter of one rectangle or for finding the perimeter of 3 rectangles or for finding the perimeter including the internal sides or for finding the perimeter excluding the two lengths of 7
	eg 3 + 10 + 3 + 10 + 3 + 10 + 3 + "7" + 10 + "7" oe **or** 4 × 10 + 4 × 3 + 2 × "7" oe **or** "78" – (4 × 3) oe **or** "72" – (2 × 3) oe **or** "52" + (2 × "7") oe			M1 for a fully correct method to find the perimeter of the shape, with at most one error (which could be one length omitted or an extra length added)
	Correct answer scores full marks (unless from obvious incorrect working)	66		A1
				Total 3 marks

Question 5

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2023 · Question 5

5	$\pi\times20\left(=20\pi=62.8(31...)\right)$ oe or $2\times\pi\times(20\div2)\left(=20\pi=62.8(31...)\right)$ oe or $0.5\times\pi\times20\left(=10\pi=31.4(15...)\right)$ oe or		3	M1 for use of $\frac{1}{2}\pi d$ or πr or πd or $2\pi r$ oe with $d = 20$ or $r = 20 \div 2$ (= 10)
	3 × ("62.8" ÷ 2) + 20 ÷ 2 + 20 ÷ 2 oe or 1.5 × ("62.8") + 20 ÷ 2 + 20 ÷ 2 oe			M1 for a complete method
	Working not required, so correct answer scores full marks (unless from obvious incorrect working)	114		A1 114 – 115 SCB1 for awrt 471
				Total 3 marks

Question 6

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · January 2021 · Question 24

Q	Working	Answer	Mark	Notes
	2 × π × 7 (= 43.982… or 14π) **or** (2 × π × 7) ÷ 2 (= 21.991… or 7π) **or** 2 × π × 9 (= 56.548… or 18π) **or** (2 × π × 9) ÷ 2 (= 28.274… or 9π)		3	M1 for finding the circumference of either the full circle or the length of the arc for either semicircle
	e.g. "21.991" + "28.274" (= 50.26…) **or** "7π" + "9π" (=16π) **or** "21.991" + "28.274" + 2 (= 52.26…) **or** "7π" + "9π" + 2 (= 52.26…) **or** "21.991" + "28.274" + 2 + 2 **or** "7π" + "9π" + 2 + 2			M1 for a method to find the length of the two arcs with intention to add
		54.3		A1 accept 54.2 – 54.3
				Total 3 marks

Question 7

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2024 · Question 13

13	$\frac{60}{360}\times2\times\pi\times r$ oe or $\frac{1}{6}\times2\times\pi\times r$ oe		3	M1 for finding the length of the arc
	"$\frac{60}{360}\times2\times\pi\times r$"$+2r$ oe or "$\frac{1}{6}\times2\times\pi\times r$"$+2r$ oe			M1 dep on M1 for a complete expression from correct working for a method for the perimeter
	Working not required, so correct answer scores full marks (unless from obvious incorrect working)	$P=r\left(\frac{1}{3}\pi+2\right)$		A1 oe eg $P=r(0.33...\pi+2)$ or $P=\left(\frac{1}{3}\pi+2\right)r$ or $P=\left(2+\frac{120}{360}\pi\right)r$ or $P=\left(\frac{120}{360}\pi+2\right)r$
				Total 3 marks

Question 8

Edexcel IGCSE Mathematics A (4MA1) · Paper 1FR · June 2025 · Question 14

Q	Working	Answer	Mark	Notes
	$6 - 3.5$ (= 2.5) **or** $4 \times 6 + 4 \times 3.5$ (= 38) **or** $6 \times 6 + 6 \times 3.5$ (= 57) or $2 \times 9.5 \times 3$ (= 57) or 19×3 (= 57)		4	M1 for a method to find the 'missing length' (may be shown on the diagram) **or** for a method to find the length of the solid lines excluding 2.5 cm, may include extra sides added **or** for a method to find the perimeter of the 3 rectangles
	$4 \times 6 + 4 \times 3.5 + 2 \times$ "2.5" (= 43) or $6 \times 6 + 6 \times 3.5 - 4 \times 3.5$ (= 43) or $2 \times 9.5 \times 3 - 4 \times 3.5$ (= 43) or $19 \times 3 - 4 \times 3.5$ (= 43)			M1 for a complete method to find the perimeter of the shape
	eg "43" × 7.6			M1 for a method to find the cost, allow use of their "43" as long as from adding at least 4 correct lengths including a length of 3.5 **and** a length of 6 eg 57 × 7.6(0) or 38 × 7.6(0)
	Correct answer scores full marks (unless from obvious incorrect working)	326.8(0)		A1
				Total 4 marks

Question 9

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · June 2022 · Question 15

Q	Working	Answer	Mark	Notes
	$48 \div 4$ (=12)		4	M1 could be on diagram
	$30 -$ "$48 \div 4$" (= 18) or 9			M1 allow 9 on correct side of the triangle on the diagram
	$3 \times$ "18" + "12" or $6 \times$ "$18 \div 2$" + "12" or "54" + "12"			M1 for a complete correct method
		66		A1
				Total 4 marks

Perimeter - Mark scheme

Question 10

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · January 2017 · Question 4

4	$\pi \times (70 - 2 \times 15)$ **or** $\pi \times 40$ (=125(.6…))		4	M1	oe
	4×15 (=60) **and** 4×70 (=280) **or** 340			M1	independent
	"125.6..." + "60" + "280"			M1	dep on M2
		466		A1	for answer in range 465.6 – 466
					Total 4 marks

Question 11

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2024 · Question 5

5	$\sqrt{81}$ (= 9) **or** 9 **or** 9×9 (= 81)		4	M1	for method to find the length of the side of the square (may be seen on the diagram)
	$4 \times$ "9" (= 36) oe			M1	for the perimeter of the square (the first M mark can be implied by 36)
	eg $\pi \times$ "9" (= 28.2(743...) or 9π)			M1	for a correct expression for the circumference for using $2\pi r$ or πD (the first M mark can be implied by 28.2(743…) rounded or truncated to 1 dp or by 9π)
	Working not required, so correct answer scores full marks (unless from obvious incorrect working)	64.3		A1	accept 64.26 – 64.3
					Total 4 marks

Question 12

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2018 · Question 6

6	$9.7^2 + 3.5^2$ (=106.34)	32.4	4	M1	M1 for the use of *MN* and a correct angle (70.1… or 70.2, 19.8…) in a correct trig statement eg $\cos 70.2 = \frac{3.5}{MN}$
	$\sqrt{9.7^2 + 3.5^2}$ or $\sqrt{"106.34"}$ (=10.3…)			M1	M1 for a complete method to find *MN* eg $MN = \frac{3.5}{\cos 70.2}$ (=10.3…)
	$\pi \times$ "10.3…" **or** $2 \times \pi \times \frac{"10.3..."}{2}$			M1	dep on M2
				A1	for answer in range 32.3 – 32.41

Question 13

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · June 2015 · Question 6

6	(a)	$1.5 \times \pi$ **or** $2 \times \pi \times (1.5 \div 2)$		2	M1
			4.71		A1 4.71 - 4.72
	(b)	$1000 \div$ "4.71 "			M1 ft from (a) (accept use of rounded answer from (a) for method mark only)
			212	2	A1 ft from (a) provided working is shown (must round down to integer value)
					Total 4 marks

Question 14

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · June 2023 · Question 20

Q	Working	Answer	Mark	Notes		
	$7.2^2 + 5.4^2 (= 81)$		4	M1	for correct first step using Pythagoras	M1 for reaching one step from the length of AB if using trig eg $(EAB =)\tan^{-1}\left(\frac{5.4}{7.2}\right)(= 36.8...)$ **and** $\sin("36.8...") = \frac{5.4}{AB}$
	$\sqrt{7.2^2 + 5.4^2}$ $(= 9)$			M1	for complete Pythagoras method to find length of AB/DC check the diagram for sight of 9, DC marked as 9 implies M2	M1 for complete method to find the length of AB/DC eg $\frac{5.4}{\sin("36.8...")}(= 9)$
	7.2 + 5.4 + 6 + "9" + 6 oe			M1	for a complete method to find the perimeter	
	Correct answer scores full marks (unless from obvious incorrect working)	33.6		A1	oe	
				Total 4 marks		

Question 15

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · January 2019 · Question 8

8	$\frac{1}{2}\times 5\times h = 12$ oe **or** $\frac{1}{2}\times 2.5\times h = 6$ oe **or** $h = 4.8$			M1	NB: 4.8 may be seen on the diagram
	$(x =)\sqrt{2.5^2 + "4.8^2"}$ **or** $(x =)\sqrt{"29.29"}$ **or** 5.41(202…)			M1	ft the candidate's value for height for this mark (award of this mark does **not** depend on award of previous mark)
	$2\times "5.41" + 5$			M1	dep on previous M1
		15.8	4	A1	for 15.8 – 15.83

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Question 16

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2023 · Question 4

4	eg $(AB^2 =)6^2+6^2(=72)$ **or** $\sin 45=\frac{6}{(AB)}$ **or** $\cos 45=\frac{6}{(AB)}$ **or** **or** $(AB^2 =)6^2+6^2-2\times6\times6\times\cos 90$		5	M1	for a correct start to the method to find AB
	eg $(AB=)\sqrt{6^2+6^2}\ (=\sqrt{72}$ or $6\sqrt{2}$ or 8.48...) **or** $(AB=)\frac{6}{\sin 45}\left(=\sqrt{72}=6\sqrt{2}=8.48...\right)$ **or** $(AB=)\frac{6}{\cos 45}\left(=\sqrt{72}=6\sqrt{2}=8.48...\right)$ **or** $(AB=)\sqrt{6^2+6^2-2\times6\times6\times\cos 90}$			M1	for a complete method to find the length of AB
	eg $\pi\times6(=6\pi$ or 18.8...) **or** $\pi\times6\div2(=3\pi$ or 9.42...) **or** $\pi\times$"8.48..."(= 26.6...) **or** $\pi\times$"8.48..."$\div 2$ (=13.3...)			M1	(indep) for a method to find the circumference of one whole circle **or** the arc length of one semicircle seen (may be embedded)
	eg $2\times$"3π"+"13.3..." **or** "9.42" + "9.42" + "13.3" **or** "18.8" + "13.3"			M1	for a complete correct method to find the perimeter of the shape
	Correct answer scores full marks (unless from obvious incorrect working)	32.2		A1	accept answers in the range 32.1 – 32.3
					Total 5 marks

Question 17

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · January 2020 · Question 11

11	$(AC^2 =)\ 17^2-15^2$		5	M1	
	$(AC=)\sqrt{17^2-15^2}\ (=\sqrt{64}=8)$			M1	
	$\frac{\pi\times'8'}{2}(=4\pi=12.566...)$			M1	dep on M2 for $\frac{\pi\times'8'}{2}$ oe or 4π 12.5663…
	'12.566…'+ 15 + 17			M1	for '12.566' + 15 + 17 and no additional values
		44.6		A1	for awrt 44.6
					Total 5 marks
Alternative mark scheme for 11					
	$\cos^{-1}\left(\frac{15}{17}\right)(=28.0724)$ or $\sin^{-1}\left(\frac{15}{17}\right)(=61.9275)$		5	M1	for a correct method to find one of the angles
	15 × tan ('28.0724')(= 8) or 15 ÷ tan ('61.9275')(= 8)			M1	
	$\frac{\pi\times'8'}{2}$ $(=4\pi=12.566…)$			M1	dep on M2 for $\frac{\pi\times'8'}{2}$ or 12.5663… or 4π
	"12.566" + 15 + 17			M1	for "12.566" + 15 + 17 and no additional values
		44.6		A1	for awrt 44.6
					Total 5 marks

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Question 18

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · January 2016 · Question 24

24	$\frac{50}{360} \times \pi \times r^2 = 20\pi$ oe		5	M1	A correct equation for area of sector
	$r^2 = \frac{360 \times 20}{50}$ or $\frac{360 \times 20\pi}{50\pi}$ oe (=144)			M1	Correct rearrangement in terms of r^2
	$r = 12$			A1	$r = 12$
	(perimeter =) $12 + 12 + \frac{50}{360} \times 2 \times 12 \times \pi$ (= $24 + \frac{10\pi}{3}$) (=24 + 10.47...)			M1ft	(ft dep on M2) for arc length $\frac{50}{360} \times 2 \times$ '12' $\times \pi$ (=10.47...) Or $l = \frac{20\pi}{12 \times 0.5}$ oe
		34.5		A1	Accept 34.4 - 34.5
					Total 5 marks

Question 19

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2024 · Question 8

8	$\frac{14+AB}{2} \times 15 = 360$ oe **or** $360 - 14 \times 15 (=150)$ oe		6	M1	for setting up an equation using the area of the trapezium **or** method to find the area of the triangle
	$AB = 34$ **or** $MB = 20$ (where M is point on AB such that MC is perpendicular to AB)			A1	could be seen on diagram
	$(CB^2 =) 15^2 + 20^2 (= 625)$ **or** $(CB^2 =) 15^2 + MB^2$			M1	allow use of their MB
	$(CB =) \sqrt{15^2 + 20^2} (= 25)$ **or** $(CB =) \sqrt{15^2 + MB^2}$			M1	allow use of their MB
	14 + 15 + "34" + "25" oe **or** 14 + 15 + 14 + MB + CB oe			M1ft	(dep on previous two M marks) for a method to find the perimeter of the trapezium, allow use of their MB and CB
	Correct answer scores full marks (unless from obvious incorrect working)	88		A1	cao
					Total 6 marks

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CHALLENGE QUESTIONS

Question 20 *Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · January 2015 · Question 24*

24	$\frac{x}{360}\times\pi\times r^2 = 5\pi$		6	M1	for this mark only condone an incorrect value for r	
	$x = 50$			A1	cao for angle $AOB = 50$	
	$(AB^2=)\ 6^2+6^2-2\times6\times6\times\cos("50")$			M1	dep on first M1 or $6\times\sin("50"/2)$	M2 for $(AB=)$ $\sqrt{25.7...}$ or 5.07....or $2\times 6\times\sin"50"/2$
	$(AB=)\ \sqrt{25.7...}$ or 5.07....			M1	dep or $2\times 6\times\sin("50"/2)$	
	$\frac{"50"}{360}\times2\times\pi\times6$ or $\frac{5\pi\times2\pi\times6}{\pi\times6^2}$ or $\frac{5}{3}\pi$ or 5.23...			M1	dep on first M1 if "50" used but indep if angle not used	
		10.3		A1	for answer in range 10.2 – 10.31	
						Total 6 marks

Question 21 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · January 2021 · Question 11*

11	$360-40\ (=320)$ or $\frac{320}{360}$ oe **or** $\frac{40}{360}\times2\pi\times9\ (=6.28...)$		4	M1	
	$\frac{"320"}{360}\times2\pi\times9\ (=16\pi=50.26...)$ **or** $2\pi\times9-"6.28"\ (=50.26)$			M1	
	"50.26" + 2 × 9			M1	complete method to find perimeter
		68.3		A1	68.2 to 68.3
					Total 4 marks

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Question 22

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · June 2026 · Question 28

Question	Working	Answer	Mark	Notes
	$\frac{1}{2}\pi r^2 = 227$ oe **or** $\pi r^2 = 227\times 2$ oe **or** $r^2 = 144.5...$ **or** $r = \sqrt{144.5...}\ (=12(.02...))$		5	M1 for a correct equation or a correct calculation to find the radius of the semicircle
	$\frac{2\pi\times"12"}{2}$ oe **or** $\pi\times"12"\ (=37.6(99...))$ **or** $\pi\times"12.02..."\ (=37.7(661...))$ **or** $\pi\times$[their radius]			M1 for a method to find the arc length of the semicircle ft their value for the radius as long as clearly identified (may be on the diagram or in the working)
	$\sqrt{15^2-"12"^2}\ (=9)$ **or** $\sqrt{15^2-"12.02..."^2}\ (=8.9(7...))$ $\sqrt{15^2-[\text{their radius}]^2}$			M1 for a complete method to find the height of the right angled triangle ft their value for the radius as long as clearly identified (may be on the diagram or in the working)
	"12"π + 2×15 + 2×"9" (= 85.6(99...)) **or** "12.02..."π + 2×15 + 2×"8.97..." (= 85.7(09...)) **or** "12"π + 48 **or** "12.02..."π + 48 **or** "37.6(99...)" + 15 + 15 + "8.97..." + "8.97..." (= 85.7(09...)) **or** [arc length] + 2×15 + 2×[*AB* or *DE*]			M1 dep on M2 for a method to find the perimeter including at least the arc length **and** one length of 15 **and** one of *AB* or *DE* ft their values for the arc length and *AB* or *DE* as long as clearly identified (may be on the diagram or in working) allow the error of addition of the radius (*OB* or OD) or the diameter *BD*
	Working required	85.7		A1 dep on M2 accept an answer in the range 85.6 – 85.8
				Total 5 marks

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Question 23

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2021 · Question 19

19	(a)	E.g. $x+y+x+y+x=100$ oe or $3x+2y=100$ oe $\left(y=\frac{100-3x}{2}\right)$		3	M1 for a correct equation for the perimeter of the shape **or** for a correct expression for the area of triangle *CED*
		E.g. $\frac{1}{2}\times x\times x\times\sin 60$ $\left(=\frac{1}{2}\times x\times x\times\frac{\sqrt{3}}{2}\right)$ $\left(=\frac{x^2\sqrt{3}}{4}\right)$	E.g. $x^2=\left(\frac{x}{2}\right)^2+h^2$ **and** $=\frac{1}{2}\times x\times\frac{x\sqrt{3}}{2}\left(=\frac{x^2\sqrt{3}}{4}\right)$		
		x"$\left(\frac{100-3x}{2}\right)$"+"$\frac{x^2\sqrt{3}}{4}$" oe			M1 for the area of the shape in terms of x only
		E.g. $x\left(\frac{200-6x}{4}\right)+\frac{x^2\sqrt{3}}{4}$ or $\frac{x}{4}(200-6x+x\sqrt{3})$ or $\frac{200x-6x^2}{4}+\frac{x^2\sqrt{3}}{4}$ or $\frac{x}{4}(200x-6x^2+x^2\sqrt{3})$	Shown		A1 for the area given in correct form with full working shown (at least one intermediate step before the answer)
	(b) (i)	$\left(\frac{dR}{dx}=\right)50-\frac{3}{2}\times 2\times x+2\times\frac{x\sqrt{3}}{4}=0$ oe		2	M1 for differentiation of correct expression with 2 out of 3 terms correct and equated to 0 (can be implied by subsequent working)
		Correct answer scores full marks (unless from obvious incorrect working)	$\frac{100}{6-\sqrt{3}}$		A1 for a correct expression
	(ii)		Correct reason	1	B1 for correct reason R is a quadratic with negative coefficient of x^2 E.g. the graph of R is ∩ shaped or (allow $\frac{d^2R}{dx^2}<0$ oe)
					Total 6 marks

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Question 24 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2025 · Question 25*

Q	Working	Answer	Mark	Notes
	eg $\frac{40}{360}\pi r^2 - \frac{1}{2}r^2 \sin 40 (= 28)$ oe **or** $\frac{40}{360}\pi r^2 = 28 + \frac{1}{2}r^2 \sin 40$ oe		6	M1 for a correct expression for the area of the shaded region Allow 3.14… or $\frac{22}{7}$ for π $\sin 40 = 0.64\ldots$
	(radius ^{2} =) 992 – 1024 (radius =) 31.8(096…)	31.8		A1 Allow answers in the range 31.5 – 32.0
	eg $(BC^2 =) 2\times"31.8"^2 - 2\times"31.8"^2 \cos 40 (= 473.4\ldots)$ **or** $\frac{0.5BC}{"31.8"} = \sin 20$ **or** $\frac{BC}{\sin 40} = \frac{"31.8"}{\sin(70)}$			M1 for a correct first step to find BC using **their clearly identified radius** eg r = …. or seen on diagram NB $\frac{180-40}{2} = 70$ $\sin 20 = 0.34\ldots$ $\sin 70 = 0.93\ldots$ or 0.94
	eg $(BC =)\sqrt{2\times"31.8"^2 - 2\times"31.8"^2 \cos 40} (= 21.7\ldots)$ **or** $(BC =) 2\times"31.8"\sin 20 (= 21.7\ldots)$ **or** $BC = \frac{"31.8"\sin 40}{\sin(70)} (= 21.7\ldots)$			M1 dep on previous M1 for a complete method to find BC $\cos 40 = 0.76\ldots$ or 0.77
	eg $\frac{130}{360}\times 2\times\pi\times"21.7"$			M1 dep on previous M1 for a complete method to find the length of arc BD
	Correct answer scores full marks (unless from obvious incorrect working)	49.4		A1 accept 48.9 – 49.7
				Total 6 marks

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